CAPSTONE THESIS TECHNICAL SUMMARY

Design of a Simple Low-Cost Repetitive Transcranial Magnetic Stimulation (rTMS) System for Low-Resource Settings

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PROJECT OVERVIEW

This undergraduate capstone project designed and modeled the power electronics, capacitor-discharge circuit topology, and magnetic stimulation coil geometry for an affordable repetitive Transcranial Magnetic Stimulation (rTMS) device tailored for major depressive disorder in resource-limited clinics.

THE PROBLEM STATEMENT

Commercial clinical rTMS machines represent major capital investments ($40,000–$100,000+) requiring specialized power conditioning and liquid chiller infrastructure. This forms an insurmountable barrier for psychiatric facilities across decentralized and developing healthcare environments.

ENGINEERING METHODOLOGY

The investigation evaluated high-voltage pulse generation using capacitor discharge topology coupled with solid-state SCR switching. Magnetic coil winding geometries (planar circular vs. figure-of-eight) were analytically modeled to evaluate cortical penetration depth against focal field concentration. Passive thermal dissipation was prioritized to eliminate costly liquid cooling loops.

SAFETY & ISOLATION ARCHITECTURE

Galvanic optocoupler isolation was incorporated between control timing circuitry and high-voltage discharge stages. Passive bleeder safety resistors and manual discharge interlocks ensured user and patient electrical safety according to IEC 60601 medical electrical principles.

ACADEMIC EVALUATION & OUTCOME

The design documentation and circuit models were successfully defended before the academic jury of the Department of Biomedical Engineering at the University of Gondar, earning Grade A distinction.